

The Predictable Tail: Historical Drawdown Predictability in a Survivorship-Conditioned Equity Panel

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1 Revision disclosure

This edition replaces the prior manuscript. It preserves the historical drawdown result within its actual sample, corrects the reported magnitude ratio, and withdraws the interpretation of payoff asymmetry as an independent return anomaly or implementable strategy. The 2009-2025 panel is survivorship-conditioned. Its observations are historical discovery evidence and are not combined with prospective TrailMap episodes.

2 Abstract

Using 715 U.S. equities observed from 2009 through 2025, this study compares the historical cross-sectional information in trailing distributional characteristics for two different targets: forward terminal return and subsequent maximum drawdown. Features are constructed point in time, but the membership universe is assembled from securities available in a later data environment. The panel is therefore survivorship-conditioned rather than a point-in-time market universe. Total volatility has a historical rank information coefficient near -0.01 for forward 60-session return and near -0.43 for realized 252-session maximum drawdown, an approximately 43-fold difference in absolute magnitude. Payoff asymmetry, idiosyncratic skewness, and MAX show the same qualitative inversion toward stronger path-risk association. The previously promoted return anomaly does not survive sector decomposition: an exact-60-session analysis gives a payoff- asymmetry spread of +6.08% per monthly formation period with Newey-West $t = 2.65$ in the full panel, but -0.27% with $t = -0.09$ after Healthcare is excluded. The corrected conclusion is descriptive: distributional characteristics merit study as historical drawdown-risk descriptors, while a return premium, causal mechanism, tradable implementation, and universe-wide generalization remain unproved.

3 1. Research question

Many equity characteristics are evaluated as predictors of average future return. That target can be poorly aligned with the practical question an investor is trying to answer. A security can end a holding period near its entry price after experiencing a large interim loss, and two securities with similar terminal returns can impose very different path risk. This paper therefore separates:

1. the rank association between a trailing characteristic and terminal forward return; and
2. the rank association between the same characteristic and peak-to-trough loss over a fixed future window.

The distinction is estimand-level, not merely statistical. A stronger association with drawdown does not automatically imply alpha, market timing, or an optimal position size. It identifies a candidate risk-description problem that requires its own prospective validation.

4 2. Data and evidence boundary

The historical panel contains 715 equities and monthly formation observations spanning 2009-2025. Trailing characteristics are computed using information available by the formation date. The return outcome is the security’s exact forward 60-session return. The risk outcome is maximum peak-to-trough drawdown during 252 sessions following formation.

The panel has four limitations that define the scope of every result:

- membership is survivorship-conditioned rather than reconstructed point in time;
- delisted firms and complete delisting returns may be underrepresented;
- repeated monthly formations for one company overlap and are not independent trials; and
- static or current classifications may not equal historical classifications.

These limitations do not make the panel useless. They make it discovery evidence. They prevent the reported coefficients from being described as market-wide prevalence, prospective confirmation, or an approval-grade investment rule.

5 3. Measurements

5.1 3.1 Distributional characteristics

The retained predictors describe different aspects of the trailing return distribution:

- total volatility measures overall dispersion;
- idiosyncratic skewness measures asymmetry after removing common variation;
- MAX records unusually large recent daily gains associated in prior literature with lottery-like demand; and
- payoff asymmetry compares the magnitude of favorable and unfavorable historical realizations.

These quantities are correlated and should not be interpreted as isolated causal mechanisms. A tail index or asymmetry statistic describes a feature of an empirical distribution under a stated window and estimator. It does not identify event origin, future drift, or the full path law.

5.2 3.2 Outcomes and clocks

Terminal return and maximum drawdown are kept separate. The outcome clock begins after the formation observation. Missing prices, corporate actions, and non-trading sessions require an explicit policy because shifting only one side of a security-benchmark comparison changes the estimand.

The prospective TrailMap design now records intended and realized dates and prices the benchmark on the security’s realized entry and exit sessions.

5.3 3.3 Dependence and inference

Monthly formations with 60- or 252-session outcomes overlap. Newey-West inference can address a specified serial-correlation structure around a specified point estimate. It cannot repair a look-ahead feature, survivor selection, sector composition, a mismatched benchmark clock, or an incomplete search ledger. The paper therefore reports effect estimates beside their limitations and treats every validation statistic as one falsification control rather than certification.

6 4. Historical drawdown result

The central surviving comparison is the much larger historical rank association with path loss:

Predictor	Forward-return rank IC	252-session drawdown rank IC	Supported interpretation
Total volatility	approximately -0.01	approximately -0.43	Stronger historical drawdown association
Lottery/distributional axis	weak and unstable	materially stronger	Candidate risk descriptor

The previous manuscript called the total-volatility contrast sixtyfold. Using the rounded values shown above, 0.43/0.01 is approximately 43. Because the denominator is close to zero, the ratio is sensitive to rounding; the coefficients should always accompany the ratio.

The result is economically interesting because maximum drawdown is path dependent. It does not show that one variable dominates every conventional risk measure, that a threshold transports to another universe, or that avoiding a high-risk group improves portfolio utility after costs.

7 5. Return result and sector falsification

The historical payoff-asymmetry return pattern appeared to survive a timing repair and overlap-aware inference. A later exact-60-session sector analysis supplied a decisive negative result:

Signal	All sectors spread / NW t	Ex-Healthcare spread / NW t
Payoff asymmetry	+6.08% / +2.65	-0.27% / -0.09
Skewness	+5.94% / +4.36	+2.07% / +1.58

Signal	All sectors spread / NW t	Ex-Healthcare spread / NW t
Tail index	-2.96% / -1.79	-0.59% / -0.41
MAX	+4.59% / +3.18	-0.19% / -0.11

For MAX, 200 seeded exclusions of the same number of randomly selected tickers produce a median spread of +3.89%; 63% retain $t > 2$. Excluding Healthcare instead produces -0.19% with $t = -0.11$. Ordinary sample-size loss is therefore not a complete explanation in that placebo design. Healthcare-only estimates are imprecise, however, so the analysis does not establish a Healthcare causal mechanism. It establishes that the aggregate relation is composition dependent.

The ex-Healthcare MAX estimate is null in each examined subperiod. The all-sector estimate is statistically clear only in 2014-2018 and the full sample. Claims of regime independence are withdrawn.

8 6. Why the prior implementation claim is withdrawn

The naive equal-weight historical long-short mean is -1.1% per formation period with $t = -0.4$. The earlier analysis capped terminal security returns and described the result as a stop-loss or risk-managed implementation. A terminal cap is a payoff sensitivity calculation, not an executable stop. It omits intraperiod paths, gap-through losses, order timing, borrow and recall, spreads, impact, option prices, portfolio capital constraints, and position availability.

A valid implementation study needs daily or finer OHLC paths, a next-observation execution rule, corporate-action treatment, borrow and option availability, costs, and a defined capital-at-risk rule. No tradability or profitability claim remains without that design.

9 7. Relationship to the TrailMap evidence system

TrailMap separates historical discovery from prospective evidence. Repeated raw detections link to canonical ticker-session-direction episodes. Each episode has separate 20-, 60-, and 120-session maturity clocks, exact security and benchmark dates, and explicit unresolved states. Historical rows do not enter prospective sample size.

This infrastructure addresses evidence accounting; it does not prove the hypothesis. A prospective test will report raw detections, canonical episodes, claim-specific clusters, sector and liquidity strata, missingness, and multiplicity-adjusted family decisions. A finding remains research-only until the predeclared effect size, false-positive, coverage, regime, and implementation gates mature.

10 8. Registered prospective design

The next test will:

1. freeze a point-in-time eligible universe, including later failures and delistings;
2. calculate characteristics only from information available at formation;
3. count canonical episodes and cluster by company and event/calendar family;
4. align security and benchmark prices on identical realized sessions;
5. evaluate 20-, 60-, and 120-session return and drawdown outcomes separately;
6. report sector, liquidity, listing-age, and regime strata;
7. retain mapping, suspension, delisting, and permanent-unresolved states; and
8. publish null and adverse outcomes with the same prominence as favorable outcomes.

11 9. Conclusion

In this survivorship-conditioned historical panel, trailing distributional characteristics carry far more rank information about subsequent drawdown than about subsequent terminal return. That asymmetry is the supported contribution. The independent payoff-asymmetry return premium, a causal Healthcare mechanism, the terminal-cap implementation, and universe-wide generalization are not supported. The negative return result is part of the paper’s conclusion rather than a hidden qualification.

12 Reproducibility statement

The source analysis and limitation record are maintained in `research/TECHNICAL_REPORT.md`, `paper/CLAIM_REGISTRY.md`, and `paper/PAPER2_SECTOR_REANALYSIS_2026-08-09.md`. The frozen exact- 60-session output is `paper/evidence/paper2_sector_exact60_2026-08-09.txt`. This edition does not pool those historical observations with prospective TrailMap data.

13 References

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14 Appendix A - Claim disposition

Prior statement	Disposition in this edition	Reason
The return anomaly is independent	Withdrawn	Aggregate result disappears ex-Healthcare
The drawdown contrast is sixtyfold	Corrected to approximately 43-fold	Arithmetic from the displayed rounded ICs
The terminal cap is a stop-loss implementation	Withdrawn	No intraperiod execution, gaps, borrow, or costs
Historical validation is prospective confirmation	Withdrawn	Historical and prospective evidence classes are separate
Distributional characteristics describe drawdown risk in this panel	Retained with limits	Fixed-horizon historical association is reproducible within the panel

15 Appendix B - Minimum replication record

A replication should preserve the eligible universe as of every formation date; security and benchmark identifiers; feature lookback endpoints; exact outcome sessions; corporate-action and delisting policies; raw rows, canonical episodes, and dependence clusters; sector and industry membership as of formation; every excluded observation and reason; code and input hashes; and the complete candidate/search ledger. Report coefficients and ratios together so a near-zero denominator cannot create a misleading headline. Historical and prospective samples must be reported in separate tables.

16 Appendix C - Interpretation guide

The paper supports use of the variables as research descriptors of future path risk. It does not support buying, shorting, excluding, or sizing a security from one characteristic. A decision rule would require incremental evidence beyond conventional risk measures, stable prospective effects, implementation costs, portfolio interactions, and an enforceable capital boundary.